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Forecasting Volatility with Alternative Data: A Hybrid GARCH–Deep Learning Framework for Sentiment and Macro–Financial Signals

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Abstract

This paper examines whether sentiment-driven information can enhance the forecasting accuracy of equity market volatility. Using daily S&P 500 returns, macroeconomic indicators from FRED, and sentiment extracted from large-scale Reddit text data, the study develops a set of econometric and deep learning models to assess the predictive role of sentiment in volatility dynamics. The empirical analysis compares a Gaussian GARCH(1,1) benchmark with a sentiment-augmented *skew-t* GARCH-X model, a feed-forward deep neural network (DNN), an LSTM sequence model, and a hybrid DNN–LSTM ensemble designed to capture both volatility clustering and nonlinear temporal structure. The results show that sentiment contributes meaningful predictive power—particularly when embedded in nonlinear learning architectures—while the *skew-t* GARCH-X specification with exogenous VIX and liquidity drivers significantly improves on the Gaussian benchmark. In the sample considered, the DNN achieves the strongest out-of-sample performance, with the hybrid and LSTM models providing complementary robustness through sequence-based dynamics. Overall, the findings demonstrate that combining econometric structure with flexible deep learning methods offers a more effective framework for volatility forecasting and underscores the value of integrating alternative sentiment data into financial econometric analysis.

Keywords: volatility forecasting, sentiment analysis, GARCH, deep learning, hybrid models, Reddit, alternative data

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1 Introduction

Forecasting stock market volatility is central to asset pricing, portfolio allocation, and risk management. Volatility exhibits persistence, clustering, and sensitivity to macro financial shocks, yet remains unobservable and must be estimated. Standard GARCH models [Engle \(1982\)](#); [Bollerslev \(1986\)](#) effectively capture volatility persistence from past returns but can struggle during rapid market changes, particularly stress episodes driven by behavioural forces. The recent surge in retail trading activity—accelerated by viral information diffusion across social media platforms such as Reddit—has reshaped U.S. equity market dynamics. Evidence from [Chicago Board Options Exchange \(2024\)](#) indicates that retail-driven sentiment is increasingly propagating into volatility and price movements. As retail participants constitute a larger share of trading volume and coordinate through online communities, their rapid shifts in sentiment introduce volatility that may elude traditional return-based models.

A substantial literature examines whether textual sentiment influences asset returns and volatility. [Tetlock \(2007\)](#) shows that pessimistic news predicts short-term return reversals and elevated trading volume. [Da et al. \(2011\)](#) construct retail attention measures from Google search data and document temporary price pressure for stocks dominated by individual investors. [Bollen et al. \(2011\)](#) find that Twitter mood variables Granger-cause broad market movements and improve neural network forecasts. More recent work integrates sentiment into machine learning frameworks, demonstrating that nonlinear models can extract predictive signals from large-scale textual data.

Despite this progress, three gaps remain. First, most studies draw sentiment from professional news or Twitter rather than retail-driven forums, which capture distinct behavioural dynamics. Second, sentiment is rarely incorporated directly into the conditional variance equations of GARCH-type models, leaving its role in volatility dynamics underexplored. Third, econometric and deep learning approaches are typically evaluated in isolation, with limited evidence on whether hybrid frameworks can improve volatility forecasts.

1.1 Research Objective and Contribution

This paper addresses these gaps by embedding Reddit-based sentiment, VIX shocks, and M2 liquidity measures into a *skew-t* GARCH-X specification, evaluating feed-forward and LSTM neural networks, and constructing a hybrid ensemble that blends econometric structure with deep learning

flexibility. Using daily S&P 500 data from January 2020 to April 2025, all models are benchmarked against a Gaussian AR(1)–GARCH(1,1) baseline to quantify the incremental value of sentiment, exogenous variables, and nonlinear modelling.

The analysis yields several key findings. Reddit-based sentiment contains measurable predictive information for S&P 500 volatility, with its contribution strongest when embedded in nonlinear models. The skew- t GARCH-X model clearly outperforms the Gaussian GARCH benchmark on information criteria and diagnostic tests, underscoring the value of exogenous variables and heavy-tailed innovations. Deep learning models deliver further improvements in out-of-sample forecast accuracy, with the feed-forward neural network achieving the strongest performance among all specifications and the hybrid ensemble providing complementary robustness by incorporating sequence-based dynamics. These findings demonstrate that sentiment-augmented nonlinear models can substantially improve volatility forecasts while motivating future work with richer sentiment sources, broader macro financial controls, and multi-horizon evaluations.

1.2 Organization of the Paper

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the data sources and variable construction. Section 4 presents the econometric and deep learning methodologies. Section 5 reports empirical results and model comparisons. Section 6 concludes and discusses implications for future research.

2 Literature Review

Research on investor sentiment in financial markets is extensive but fragmented. Early work focused on professional news sources, followed by studies using online search-based measures of retail attention, and more recent contributions analyzing real-time sentiment from social media. Despite abundant research on sentiment and asset returns, far less is known about sentiment’s contribution to *volatility forecasting*, and existing studies seldom integrate sentiment directly into *conditional variance models* such as GARCH. Moreover, econometric and deep learning approaches are typically evaluated separately, leaving open the question of whether hybrid frameworks can improve forecast performance.

2.1 News-Based Sentiment

The first strand examines news-based sentiment and its effects on returns and trading activity. [Tetlock \(2007\)](#), for example, constructs a daily sentiment index from Wall Street Journal columns and shows that spikes in pessimism predict short-term return reversals and heightened trading volume, highlighting a measurable behavioral component in market dynamics. While the GARCH framework of [Engle \(1982\)](#) and [Bollerslev \(1986\)](#) has become standard for modelling volatility clustering, these models typically omit behavioral variables. Although influential, this literature focuses on return predictability rather than volatility and relies on professional media sources rather than retail-driven sentiment.

2.2 Search-Based Attention Measures

A second strand studies *investor attention* using search data. [Da et al. \(2011\)](#) introduce the Google Search Volume Index (SVI) and develop a stationary abnormal search-volume metric that predicts temporary price pressure and subsequent reversals. Their findings reveal that retail-driven attention shocks contain forecastable structure, particularly for stocks dominated by individual investors. This work shows retail sentiment can be transformed into a usable time-series factor but remains oriented toward returns rather than conditional volatility.

2.3 Social Media-Driven Sentiment

A third strand incorporates *real-time social media sentiment*. [Bollen et al. \(2011\)](#) demonstrate that Twitter mood variables—especially the “Calm” dimension—Granger-cause broad market movements and enhance neural-network forecasts of DJIA direction. Their results suggest nonlinear models can extract predictive signals from large-scale sentiment data, although the evidence is sample-specific and not directly applied to volatility forecasting.

2.4 Market Structure and Retail Impact

Complementary evidence highlights the growing relevance of retail-driven sentiment. Industry reports show retail traders now account for a structurally higher share of U.S. equity volume, facilitated by commission-free platforms and viral propagation of information on social media. These developments imply that sentiment shocks originating from retail platforms—such as Reddit—may contribute more strongly to market volatility than in previous decades.

2.5 Synthesis and Research Gaps

Taken together, the literature shows that (i) sentiment and attention measures affect market behavior, (ii) retail-driven sentiment is increasingly influential, and (iii) nonlinear models can extract predictive signals beyond traditional econometric methods. However, three gaps remain. First, sentiment is rarely incorporated directly into GARCH-type variance equations, leaving its role in volatility dynamics an open question. Second, few studies evaluate sentiment within deep learning architectures alongside macro-financial controls. Third, limited research systematically compares GARCH, deep learning, and hybrid ensemble approaches for forecasting volatility.

This paper addresses these gaps in three ways. First, it embeds Reddit sentiment, VIX shocks, and macro-liquidity measures into a *skew-t* GARCH-X model. Second, it evaluates their nonlinear interactions using feedforward and LSTM architectures. Third, it constructs a hybrid ensemble to test whether combining econometric structure with deep learning yields superior volatility forecasts.

3 Data

This study integrates (i) financial market data, (ii) Reddit-derived sentiment indicators, (iii) macroeconomic series from FRED, and (iv) custom exogenous variables to construct a dataset suitable for econometric and deep learning volatility forecasting. All datasets were aligned and transformed to ensure stationarity and comparability across series. The final sample spans January 2020 to April 2024.

3.1 Financial Market Data

The primary financial variable is the daily logarithmic return of the S&P 500 Index (SPX), sourced from [Yahoo Finance \(2025\)](#):

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (1)$$

This return series provides a high-frequency measure of equity market behaviour and serves as the target variable for volatility modelling. Additional market-based indicators were collected from [Yahoo Finance \(2025\)](#) to support feature engineering and robustness checks:

- **SPDR S&P 500 ETF (SPY):** A highly liquid ETF tracking the S&P 500.
- **CBOE Volatility Index (VIX):** Measures implied volatility in S&P 500 options and acts as a gauge of market uncertainty [Whaley \(2000\)](#).
- **10-Year Treasury Yield (TNX):** Captures long-term interest rate conditions and the opportunity cost of holding equities.

These variables contribute complementary information on volatility, market stress, liquidity conditions, and investor risk perception. Time series plots of these variables and their corresponding returns are presented in Figures 1–8 in the Appendix. The correlation structure among key asset classes is illustrated in Figure 9.

3.2 Reddit Sentiment Data

Reddit posts were retrieved using the PRAW API [Praw Development Team \(2023\)](#) from seven high-activity finance subreddits: r/WallStreetBets, r/Investing, r/Stocks, r/SecurityAnalysis, r/Economics,

r/Options, and r/Trading. Posts mentioning key tickers (SPX, SPY, VIX, TNX, QQQ, and major “Magnificent Seven” firms) were retained, yielding 3,900 unique observations with metadata (subreddit, timestamp, score, and text).

Noise reduction filters excluded posts containing profanity, speculative slang (e.g., “YOLO”, “diamond hands”), or near-duplicate submissions by the same user within 24 hours.

Text preprocessing included:

- lowercasing;
- stop-word removal using NLTK [Bird et al. \(2009\)](#);
- lemmatization using spaCy [Honnibal et al. \(2020\)](#);
- removal of URLs, tickers, hashtags, and special characters.

Sentiment was quantified using VADER compound scores [Hutto and Gilbert \(2014\)](#) and TextBlob polarity to ensure robustness to finance-specific language. Engagement-weighted scores were constructed using $\ln(1 + \text{score} + \text{comments})$, and daily averages were shifted forward by one day to evaluate predictive sentiment spillovers:

$$\text{WeightedSentiment}_t = s_t \cdot \ln(1 + \text{score} + \text{comments}) \quad (2)$$

The time series of daily weighted sentiment and its distribution are shown in Figures 11 and 12, respectively.

3.3 Macroeconomic Data

Macroeconomic series were obtained from the Federal Reserve Economic Data (FRED) database [Federal Reserve Bank of St. Louis \(2025\)](#), which aggregates U.S. and international macroeconomic indicators. The selected series reflect monetary policy, inflationary pressure, labour market conditions, and real economic activity.

First-differencing was applied to bounded variables (e.g., interest rates, unemployment), while trending aggregates were log-differenced to stabilize variance. All macro series were resampled to daily frequency using forward-fill to synchronize with financial data while avoiding any forward-

Table 1: Macroeconomic Variables from FRED

Series	FRED Code	Frequency	Transformation
Federal Funds Rate	FEDFUNDS	Monthly	First difference
Unemployment Rate	UNRATE	Monthly	First difference
Consumer Price Index (CPI)	CPIAUCSL	Monthly	Log difference
M2 Money Supply	M2SL	Monthly	Log difference
Real GDP	GDPC1	Quarterly	Log difference
Personal Consumption	PCECC96	Monthly	Log difference
Nonfarm Payroll Employment	PAYEMS	Monthly	Log difference

looking bias. The evolution of key macroeconomic indicators over the sample period is displayed in Figure 10.

3.4 Exogenous Feature Construction

A broad set of candidate variables was evaluated using Granger causality tests Granger (1969). The final exogenous predictors and their lag structures are presented in Table 2. Detailed Granger causality test results are provided in Figure 23 in the Appendix.

Table 2: Exogenous Features and Lag Structure

Variable	Lag	Rationale
VIX Returns	2	Captures short-term volatility shocks
M2 Money Supply Growth	2	Reflects liquidity-driven risk conditions
Weighted Reddit Sentiment	5	Incorporates delayed behavioural responses

Short-term realized measures—including 3-day realized variance and the 3-day sum of absolute returns—were also included to capture volatility clustering and market microstructure effects. Autocorrelation functions for key variables are presented in Figures 13–18.

3.5 Final Dataset Assembly

All variables were merged into a unified dataset, `aligned_data`, indexed by trading date. Observations affected by lagging, rolling windows, or missing values were removed to maintain consistency across models.

The final dataset contains:

- S&P 500 log returns (target series);

- lagged exogenous predictors (VIX returns, M2 growth, Reddit sentiment);
- realized volatility proxies;
- transformed macroeconomic variables;
- GARCH-derived conditional volatility and standardized residuals.

The target series is stored as `aligned_returns`, while all predictors are housed in `exog_features`. This structure supports both traditional econometric models and deep learning architectures.

4 Empirical Approach

This study develops a multi-model forecasting framework that combines traditional econometric volatility models with deep learning architectures. All models are estimated on data from January 2020 to December 2023 and evaluated on a holdout period from January to April 2024. The methodological pipeline consists of four components: (i) a baseline AR(1)–GARCH(1,1) model, (ii) a sentiment-augmented ARX(1)–GARCH-X(1,1) specification with *skewed Student-t* innovations, (iii) nonlinear deep learning models (DNN and LSTM), and (iv) a hybrid ensemble that blends econometric structure with sequential deep learning.

4.1 Baseline Model: AR(1)–GARCH(1,1)

The baseline specification is an AR(1)–GARCH(1,1) framework, a canonical model in financial econometrics [Engle \(1982\)](#); [Bollerslev \(1986\)](#) for capturing equity return and volatility dynamics. The AR(1) component in the mean equation accounts for short-run autocorrelation in log returns:

$$r_t = \mu + \phi r_{t-1} + \varepsilon_t, \quad (3)$$

where r_t denotes S&P 500 log returns and ε_t are mean-zero innovations.

Volatility clustering and persistence are modelled using a GARCH(1,1) variance equation:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (4)$$

where ω is the constant term, α measures sensitivity to recent shocks (ARCH effect), and β captures

volatility persistence. The parameters satisfy the usual stationarity and positivity constraints,

$$\omega > 0, \quad \alpha \geq 0, \quad \beta \geq 0, \quad \alpha + \beta < 1, \quad (5)$$

ensuring a finite unconditional variance and a well-behaved process. Standardized innovations are assumed Gaussian, and this model serves as the benchmark for assessing the incremental value of exogenous information and nonlinear structures. Model estimation results and conditional volatility are presented in Figures 19 and 20.

4.2 Sentiment-Augmented ARX(1)–GARCH-X(1,1)

To incorporate behavioural, macroeconomic, and market-based information, exogenous predictors are added to the AR(1)–GARCH(1,1) benchmark, yielding an ARX(1)–GARCH-X(1,1) specification Creal et al. (2013). Lagged VIX returns, M2 money growth, and Reddit-based sentiment enter both the conditional mean and variance equations. Granger causality tests Granger (1969) guide the selection of informative lags.

Mean equation:

$$r_t = \mu + \phi r_{t-1} + \beta_1 \Delta \text{VIX}_{t-1} + \beta_2 \Delta \text{M2}_{t-2} + \beta_3 \text{Sentiment}_{t-3} + \varepsilon_t. \quad (6)$$

Variance equation:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta_1 \Delta \text{VIX}_{t-2} + \delta_2 \Delta \text{M2}_{t-2} + \delta_3 \text{Sentiment}_{t-5}. \quad (7)$$

Given the asymmetric heavy tails of equity-index returns, the GARCH-X model is estimated under a *skewed Student-t* innovation distribution. This accommodates return asymmetry and excess kurtosis and aligns with established volatility-modelling practice. A comparison of innovation distributions (*Normal*, *Student-t*, and *Skew-t*) based on information criteria is shown in Figure 21, with conditional volatility estimates across specifications presented in Figure 22.

4.3 Deep Neural Network (DNN)

A feedforward neural network models nonlinear relationships between exogenous predictors and next-day volatility. All predictors—including lagged VIX returns, lagged M2 growth, engagement-weighted Reddit sentiment, realized volatility proxies (3-day realized variance and sum of absolute returns), and GARCH-derived features—are standardized prior to training along with the target (next-day log-volatility).

4.3.1 Network Architecture

- **Input layer:** dimension equals the number of standardized predictors;
- **Hidden layer 1:** 32 ReLU units, $h_1 = \text{ReLU}(W_1x + b_1)$;
- **Hidden layer 2:** 16 ReLU units, $h_2 = \text{ReLU}(W_2h_1 + b_2)$;
- **Output layer:** 1 linear unit, $\hat{y}_{t+1} = W_3h_2 + b_3$.

4.3.2 Training Procedure

- Loss: mean squared error (MSE);
- Optimizer: Adam ($\alpha = 0.001$);
- Batch size: 32;
- Epochs: 50;
- Data split: chronological 60% training, 20% validation, 20% testing.

The model minimizes:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (8)$$

The DNN captures nonlinear cross-sectional interactions among sentiment, macroeconomic, and market-based predictors but does not explicitly model temporal dependence. Feature importance analysis using SHAP values is provided in Figure 24, while residual diagnostics are shown in Figure 27.

4.4 Long Short-Term Memory (LSTM) Model

To capture sequential structure, temporal persistence, and nonlinear dynamics, an LSTM architecture is implemented [Hochreiter and Schmidhuber \(1997\)](#). LSTMs retain information across time via gating mechanisms, making them well-suited to volatility forecasting.

4.4.1 Sequential Construction

A sliding window of length five constructs sequential inputs:

$$X_t = [x_{t-5}, x_{t-4}, x_{t-3}, x_{t-2}, x_{t-1}], \quad (9)$$

and the target is next-day log-volatility:

$$y_t = \log(1 + \sigma_t^2), \quad (10)$$

which stabilizes variance and mitigates the influence of heavy tails.

4.4.2 Network Architecture

- Gaussian noise layer for robustness;
- LSTM layer 1: 128 units with L2 regularization, `return_sequences = True`;
- Dropout (10%);
- LSTM layer 2: 64 units with batch normalization;
- Dense layer: 32 ReLU units with 5% dropout;
- Output layer: 1 linear neuron.

4.4.3 Weighted Huber Loss

Prediction error:

$$e_t = y_t - \hat{y}_t, \quad (11)$$

with Huber loss:

$$L_{\delta}(e_t) = \begin{cases} \frac{1}{2}e_t^2, & |e_t| \leq \delta, \\ \delta|e_t| - \frac{1}{2}\delta^2, & |e_t| > \delta. \end{cases} \quad (12)$$

Observation-specific weights increase with realized volatility, placing greater weight on periods of heightened market stress.

4.4.4 Training Procedure

- Target: next-day log-volatility $y_t = \log(1 + \sigma_t^2)$;
- Loss: weighted Huber;
- Optimizer: Adam with cosine-decay restarts;
- Learning rate: 2×10^{-3} ;
- Epochs: 80;
- Batch size: 16;
- Early stopping: patience 15, based on validation MAE;
- Data split: chronological 60% training, 20% validation, 20% test.

Residual diagnostics for the LSTM model are presented in Figure 28.

4.5 Hybrid DNN–LSTM Ensemble

The hybrid model is motivated by the observation that econometric models excel at capturing long-memory variance persistence, while deep learning architectures capture nonlinear and sequential dynamics that traditional models cannot. By aggregating forecasts, the ensemble leverages complementary strengths: GARCH-based features encode persistent volatility structure, the DNN captures nonlinear cross-sectional interactions, and the LSTM models sequential patterns and volatility clustering.

4.5.1 Hybrid Forecast

$$\hat{\sigma}_{\text{Hybrid},t+1}^2 = \alpha \hat{\sigma}_{\text{LSTM},t+1}^2 + (1 - \alpha) \hat{\sigma}_{\text{DNN},t+1}^2, \quad (13)$$

where $\alpha \in [0, 1]$ controls the weighting.

4.5.2 Weight Grid and Selection

Weights are searched over a fine grid:

$$\alpha \in \{0, 0.02, 0.04, \dots, 1.00\}, \quad (14)$$

and the optimal weight minimizes validation MAE:

$$\alpha^* = \arg \min_{\alpha} \text{MAE}(\sigma_{t+1}^2, \alpha \hat{\sigma}_{\text{LSTM},t+1}^2 + (1 - \alpha) \hat{\sigma}_{\text{DNN},t+1}^2). \quad (15)$$

4.5.3 Re-Scaling of Forecasts

Because the models are trained on standardized log-volatility, final forecasts are de-scaled in two steps:

1. Apply the inverse standardization transformation to recover $\tilde{y}_t$;
2. Recover variance via $\sigma_t^2 = \exp(\tilde{y}_t) - 1$, where $\tilde{y}_t$ is the de-standardized log-volatility.

This ensures comparability across models and interpretable volatility forecasts on the original scale. Residual diagnostics for the hybrid ensemble are shown in Figure 29, while comparative performance across all deep learning models is presented in Figures 25, 26, and 30.

5 Results

5.1 AR(1)–GARCH(1,1) Baseline Results

The analysis begins with a baseline AR(1)–GARCH(1,1) model estimated on daily S&P 500 returns under a conditional Normal distribution. This specification serves as a parsimonious benchmark that captures both short-run return dependence and time-varying volatility.

Table 3: Distribution Comparison for AR(1)–GARCH(1,1)

Distribution	AIC	BIC	LogLik	Converged
Normal	−8236.499709	−8210.587949	4123.249855	True
Skew- t	3296.563112	3332.839577	−1641.281556	True
Student- t	8038.751583	8069.845696	−4013.375792	True

The estimated variance dynamics satisfy the usual stationarity and positivity conditions, with a highly persistent conditional variance process: the sum of the ARCH and GARCH coefficients is close to unity ($\alpha + \beta \approx 0.98$), consistent with the slow-moving volatility typically observed in equity index data. The resulting conditional volatility series displays pronounced volatility clustering, sharp spikes around periods of market stress, and gradual mean reversion (Figure 20). Standardized residuals are approximately mean-zero, exhibit no remaining serial correlation or residual ARCH effects (Ljung–Box and ARCH–LM diagnostics), and show substantially reduced heteroscedasticity relative to raw returns. Although the AR(1) term in the mean equation is statistically significant, the model explains only a small fraction of return variation, underscoring the limited predictability of equity returns using linear structure alone. Full model estimation results are presented in Figure 19.

Overall, the Gaussian AR(1)–GARCH(1,1) provides a statistically adequate volatility benchmark, but its reliance solely on lagged returns and volatility motivates the exploration of richer models with exogenous information and more flexible innovation distributions.

5.2 Sentiment-Augmented GARCH-X Model Results

The GARCH-X(1,1) specification is estimated in R under three innovation *distributions*—*Normal*, *Student-t*, and *skew-t*—using lagged VIX returns (lag 2), M2 money growth (lag 2), and Reddit-based sentiment (lag 5) as exogenous regressors in both the conditional mean and variance equations.

Model Selection

Table 4: GARCH Model Selection: Innovation Distributions

Distribution	AIC/obs	BIC/obs	LogLik
Skew- t (sstd)	−6.4728	−6.3996	2734.81
Student- t	−6.4592	−6.3916	2728.07
Normal	−6.4148	−6.3529	2708.42

Model comparison via AIC, BIC, and log-likelihood strongly favours the *skew-t* specification (Table 4). The skewness parameter is highly significant ($t \approx 19$), confirming that the return distribution is asymmetric with heavy tails—properties that standard Normal innovations cannot capture. A visual comparison of innovation distributions and their impact on conditional volatility estimates is provided in Figures 21 and 22.

Because the baseline AR(1)–GARCH(1,1) model is estimated in Python (which reports total AIC/BIC), whereas the R implementation reports per-observation criteria, the baseline values are divided by sample size. After harmonization, the *skew-t* GARCH-X(1,1) remains superior: the baseline achieves $\text{AIC} \approx -6.26$ and $\text{BIC} \approx -6.24$ per observation, while the *skew-t* model attains $\text{AIC} \approx -6.47$ and $\text{BIC} \approx -6.40$.

Interpretation

Several findings emerge:

- The GARCH-X specification improves on the baseline by incorporating market volatility (VIX), liquidity conditions (M2), and sentiment shocks. These covariates enter with economically meaningful signs under the *skew-t* distribution and materially enhance overall fit.
- Volatility persistence remains very high ($\alpha + \beta \approx 0.986$), consistent with long-memory behavior typical of equity-index volatility.
- The *skew-t* specification produces smoother conditional volatility during stress episodes, reflecting its ability to capture asymmetric, heavy-tailed innovations.
- Linear sentiment effects are modest ($t \approx 0.19$), suggesting that sentiment’s role is nonlinear and better captured by deep learning models.
- Residual diagnostics (Ljung–Box, squared residual tests, ARCH–LM) show no remaining serial correlation or ARCH effects, confirming statistical adequacy.

Overall, the *skew-t* GARCH-X(1,1) model outperforms the Gaussian benchmark, offering more accurate and interpretable volatility estimates while remaining parametric and linear in its treatment of exogenous variables.

5.3 Deep Neural Network (DNN) Model Results

The feedforward multilayer perceptron forecasts next-day log conditional volatility from standardized exogenous predictors (GARCH conditional variance, residuals, VIX returns, M2 growth, Reddit sentiment, and realized-volatility proxies). The dataset is partitioned chronologically into 499 training observations (59.9%), 167 validation observations (20%), and 167 test observations (20%). The network, trained using Adam ($\alpha = 0.001$) with early stopping, converges robustly.

On the test set, the DNN attains $\text{MAE} = 0.2169$, $\text{MSE} = 0.0811$, and $R^2 = 0.8673$ on scaled log-volatility. After de-scaling, this corresponds to a real-space MAE of approximately 6.9×10^{-4} in daily variance units. Training and validation losses decline rapidly within the first 10–15 epochs and then stabilize, with no signs of overfitting.

Residual diagnostics indicate forecast errors tightly centered around zero, with skewness = 0.40, kurtosis = 3.94, and Shapiro–Wilk $p = 0.047$ (Figure 27). SHAP-based importance scores suggest the DNN relies primarily on GARCH conditional volatility and standardized residual shocks, with secondary contributions from VIX and sentiment (Figure 24).

Relative to *skew-t* GARCH-X, the DNN delivers substantially lower out-of-sample MAE and higher R^2 . Although GARCH-X fits well in sample, it exhibits sign bias ($p = 0.022$) and heavy-tailed residuals (shape ≈ 11.47), limiting generalization. The DNN eliminates sign bias and captures nonlinear interactions among volatility, macro-liquidity, and sentiment that GARCH-X cannot encode.

5.4 LSTM Sequence Model Results

The LSTM is trained on 5-day sequences of volatility-related predictors using the same chronological 60/20/20 split. On the test set, it achieves $\text{MAE} = 0.3266$, $\text{MSE} = 0.1950$, and $R^2 = 0.6866$ on scaled log-volatility, corresponding to a real-space MAE of approximately 9.6×10^{-4} .

Training and validation MAE decline rapidly and then plateau, with no evidence of overfitting. Out-of-sample predictions track realized volatility well, capturing peaks and troughs with modest delay.

Residuals are near mean-zero (skewness = 0.135, kurtosis = 3.70, Shapiro–Wilk $p \approx 2.2 \times 10^{-5}$), as shown in Figure 28. The residual distribution is tighter than that of the *skew-t* GARCH-X model,

reflecting the LSTM’s ability to capture nonlinear temporal structure. Although less accurate than the DNN, it substantially outperforms *skew*-t GARCH-X out of sample.

5.5 Hybrid DNN–LSTM Ensemble Results

A linear ensemble combines the DNN and LSTM forecasts:

$$\hat{\sigma}_{t+1,\text{Hybrid}}^2 = \alpha \hat{\sigma}_{t+1,\text{LSTM}}^2 + (1 - \alpha) \hat{\sigma}_{t+1,\text{DNN}}^2,$$

with $\alpha \in [0, 1]$ selected via grid search over

$$\alpha \in \{0, 0.02, 0.04, \dots, 1.00\}.$$

The optimal weight is $\alpha^* = 0.46$.

On the test set, the ensemble achieves $\text{MAE} = 0.000696$, $\text{MSE} = 0.000001$, $\text{RMSE} = 0.000938$, and $R^2 = 0.8480$ in real-scale variance units. Residuals are centered near zero with skewness $= 0.0368$, kurtosis $= 2.17$, and Shapiro–Wilk $p \approx 3.5 \times 10^{-3}$, displaying well-behaved tails and minimal asymmetry (Figure 29).

The hybrid retains most of the DNN’s predictive power while smoothing fluctuations arising from the LSTM. It provides a robustness-oriented complement to the DNN, trading minimal accuracy loss for highly stable residual diagnostics.

5.6 Final Comparative Assessment

Table 5: Model Comparison Summary (Test Set)

Model	Type	AIC/obs	BIC/obs	$\alpha + \beta$	MAE	MSE	RMSE	R^2
AR(1)–GARCH(1,1)	Gaussian baseline	−6.26	−6.24	0.98	–	–	–	–
ARX–GARCH-X	skew- <i>t</i> , VIX, M2, Senti- ment	−6.47	−6.40	0.99	–	–	–	–
DNN	nonlinear cross-section	–	–	–	0.2169	0.0811	0.2848	0.8673
LSTM	5-day sequences	–	–	–	0.3266	0.1950	0.4416	0.6866
Hybrid ($\alpha = 0.46$)	46% LSTM + 54% DNN	–	–	–	0.000696 [†]	0.000001 [†]	0.000938 [†]	0.8480

Notes: AIC/BIC are reported per observation for GARCH models. DNN and LSTM metrics are on scaled log-volatility.

[†]Hybrid MAE/MSE/RMSE are in real-scale variance units; scaled values are used for R^2 .

Overall, the econometric benchmarks provide disciplined reference points. On a per-observation basis, the *skew*-t ARX–GARCH-X(1,1) clearly dominates the Gaussian GARCH model, underscor-

ing the value of exogenous volatility, liquidity, and sentiment drivers combined with heavy-tailed innovations. Once nonlinear architectures exploiting the same inputs are introduced, however, the largest forecasting gains arise from the deep learning models: the DNN achieves the strongest out-of-sample accuracy, the LSTM adds complementary sequence-based dependence, and the hybrid ensemble offers a robust accuracy–stability trade-off. Comparative forecasts across all deep learning models are visualized in Figures 25, 26, and 30. Together, the results indicate that sentiment and macro-augmented nonlinear models provide the most effective framework for short-horizon volatility forecasting.

6 Conclusion

6.1 Summary of Findings

This paper evaluated whether sentiment-driven information can improve short-horizon volatility forecasts for the S&P 500. Using Reddit-based sentiment, macrofinancial indicators, and a suite of econometric and deep learning models, the analysis compared a Gaussian AR(1)–GARCH(1,1) benchmark with a *skew-t* GARCH-X specification, a feedforward DNN, an LSTM sequence model, and a hybrid ensemble.

On a per-observation basis, the *skew-t* GARCH-X model substantially outperformed the Gaussian baseline, confirming that heavy-tailed, asymmetric innovations and exogenous VIX and liquidity effects are essential for capturing equity-index volatility. Among nonlinear models, the DNN achieved the strongest out-of-sample accuracy, while the LSTM added complementary sequence dependence. The hybrid ensemble delivered accuracy close to the DNN with particularly stable residual behaviour.

Overall, the findings show that sentiment has only modest linear effects in a GARCH-X specification but becomes economically meaningful when embedded in flexible nonlinear architectures. Integrating behavioural signals, macrofinancial covariates, and deep learning produces a more powerful and robust framework for forecasting equity-market volatility.

6.2 Limitations

Several limitations warrant caution. First, the sample size is modest and relies on a single sentiment source (Reddit), which may not reflect broader investor behaviour. Second, the macrofinancial covariates are restricted to VIX and M2, and the performance of the GARCH-X model is conditional on the *skew-t* innovation assumption. Third, the deep learning models are estimated on a relatively small dataset, making their comparative performance sensitive to the sample period, architectural choices, and prevailing market conditions. Distinct market regimes—such as prolonged stress or extended calm—could therefore yield different rankings of the DNN, LSTM, and hybrid models.

6.3 Directions for Future Research

Future work could incorporate high-frequency realized volatility to better capture intraday dynamics and expand sentiment beyond Reddit to include financial news, Twitter, prediction markets, and multi-platform behavioural indices. Enriching the macrofinancial information set with credit spreads, interest-rate volatility, liquidity indicators, and option-implied measures would also improve robustness. Finally, evaluating performance across multiple forecast horizons (1-day, 5-day, and 22-day) and distinct market regimes would clarify how the relative strengths of DNNs, LSTMs, and hybrid ensembles evolve through time and whether sentiment-driven nonlinear models retain their advantage under different economic environments.

7 Acknowledgements

This paper originated as coursework for ECON 423 at the University of Waterloo. I am grateful to Professor Dinghai Xu for his guidance. All errors remain my own.

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Appendix

List of Tables

1	Macroeconomic Variables from FRED	10
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4	GARCH Model Selection: Innovation Distributions	17
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List of Key Equations

Logarithmic S&P 500 return

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (16)$$

Engagement-weighted Reddit sentiment

$$\text{WeightedSentiment}_t = s_t \cdot \ln(1 + \text{score} + \text{comments}) \quad (17)$$

AR(1) mean equation

$$r_t = \mu + \phi r_{t-1} + \varepsilon_t \quad (18)$$

GARCH(1,1) variance equation

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (19)$$

GARCH(1,1) parameter constraints

$$\omega > 0, \quad \alpha \geq 0, \quad \beta \geq 0, \quad \alpha + \beta < 1 \quad (20)$$

ARX(1) mean equation

$$r_t = \mu + \phi r_{t-1} + \beta_1 \Delta \text{VIX}_{t-1} + \beta_2 \Delta \text{M2}_{t-2} + \beta_3 \text{Sentiment}_{t-3} + \varepsilon_t \quad (21)$$

GARCH-X(1,1) variance equation

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta_1 \Delta \text{VIX}_{t-2} + \delta_2 \Delta \text{M2}_{t-2} + \delta_3 \text{Sentiment}_{t-5} \quad (22)$$

DNN hidden layers and output

$$h_1 = \text{ReLU}(W_1 x + b_1), \quad h_2 = \text{ReLU}(W_2 h_1 + b_2), \quad \hat{y}_{t+1} = W_3 h_2 + b_3 \quad (23)$$

DNN loss (MSE)

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (24)$$

Next-day log-volatility transform

$$y_t = \log(1 + \sigma_t^2) \quad (25)$$

Prediction error for Huber loss

$$e_t = y_t - \hat{y}_t \quad (26)$$

Huber loss

$$L_\delta(e_t) = \begin{cases} \frac{1}{2}e_t^2, & |e_t| \leq \delta, \\ \delta|e_t| - \frac{1}{2}\delta^2, & |e_t| > \delta \end{cases} \quad (27)$$

Hybrid forecast definition

$$\hat{\sigma}_{\text{Hybrid},t+1}^2 = \alpha \hat{\sigma}_{\text{LSTM},t+1}^2 + (1 - \alpha) \hat{\sigma}_{\text{DNN},t+1}^2, \quad \alpha \in [0, 1] \quad (28)$$

Hybrid weight grid

$$\alpha \in \{0, 0.02, 0.04, \dots, 1.00\} \quad (29)$$

Optimal hybrid weight

$$\alpha^* = \arg \min_{\alpha} \text{MAE}(\sigma_{t+1}^2, \alpha \hat{\sigma}_{\text{LSTM},t+1}^2 + (1 - \alpha) \hat{\sigma}_{\text{DNN},t+1}^2) \quad (30)$$

List of Key Figures

Data and Descriptive Figures



Figure 1: S&P 500 Index Price Series



Figure 2: SPY ETF Price Series

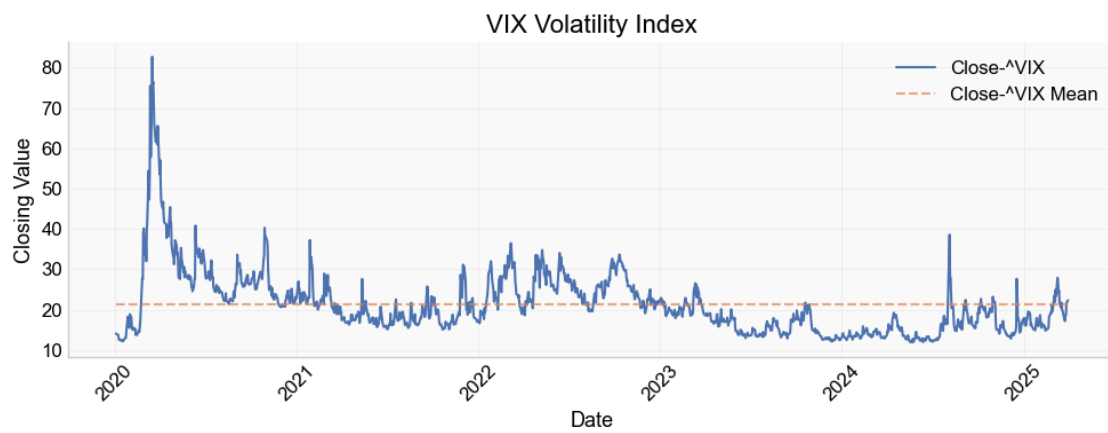


Figure 3: VIX Index Time Series



Figure 4: 10-Year Treasury Yield (TNX)

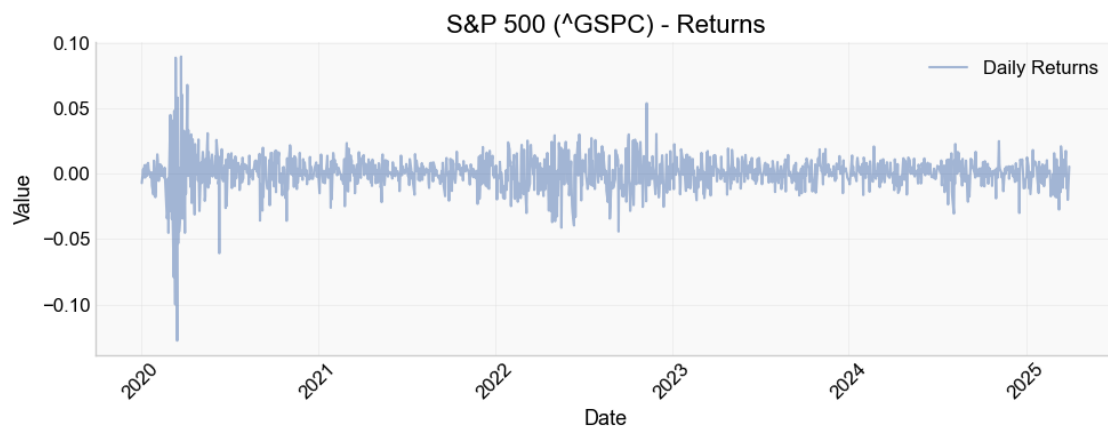


Figure 5: Daily S&P 500 Returns

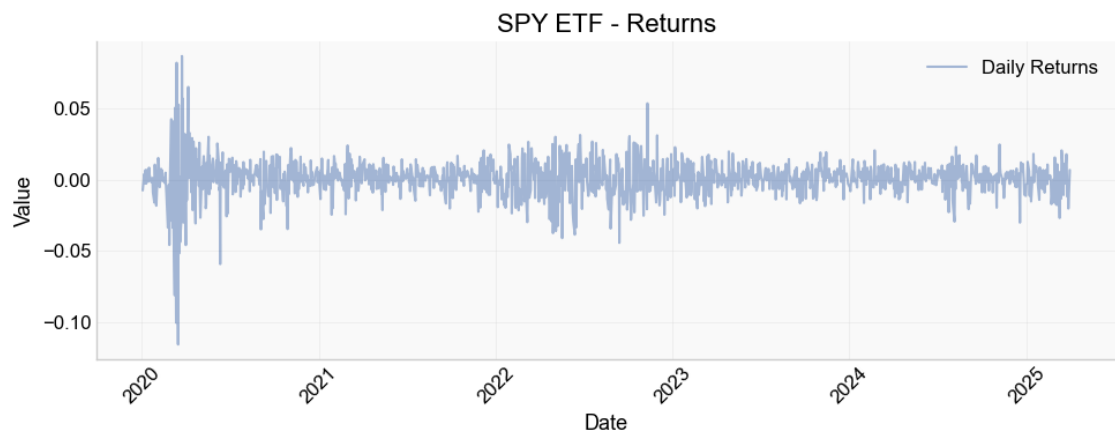


Figure 6: Daily SPY Returns

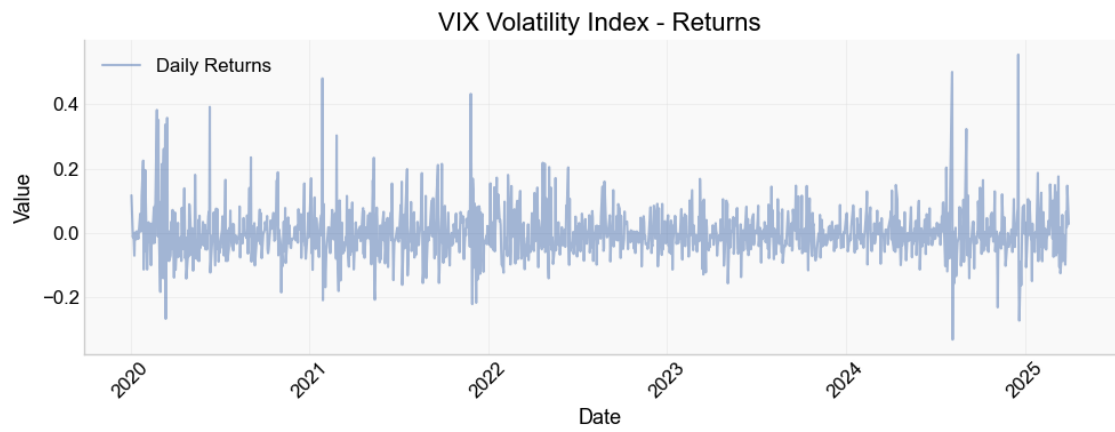


Figure 7: Daily VIX Returns

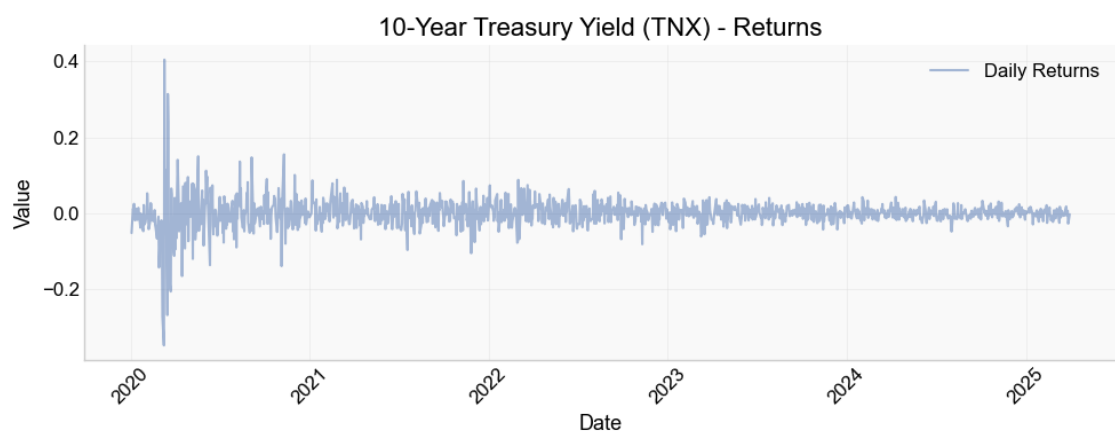


Figure 8: Daily TNX Returns

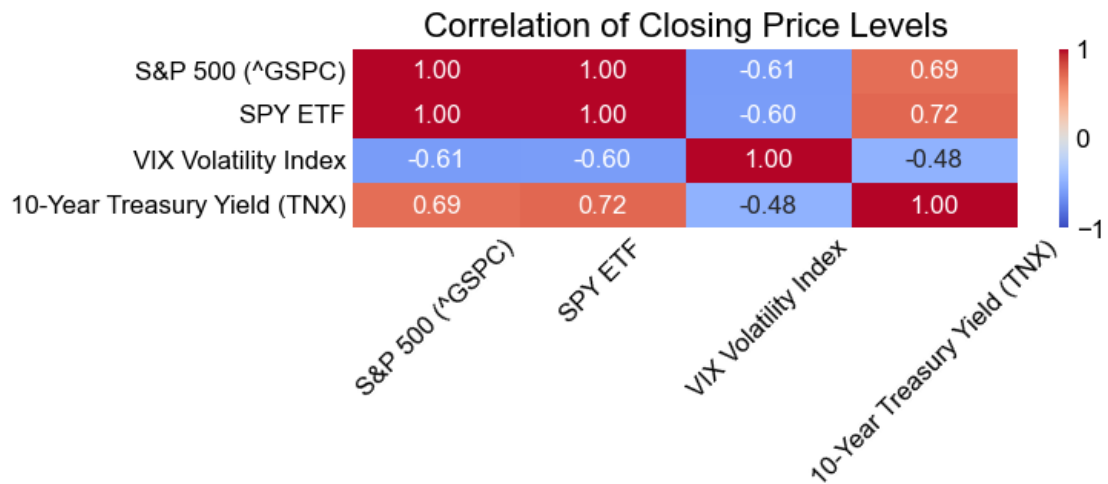


Figure 9: Correlation Matrix of Key Asset Classes

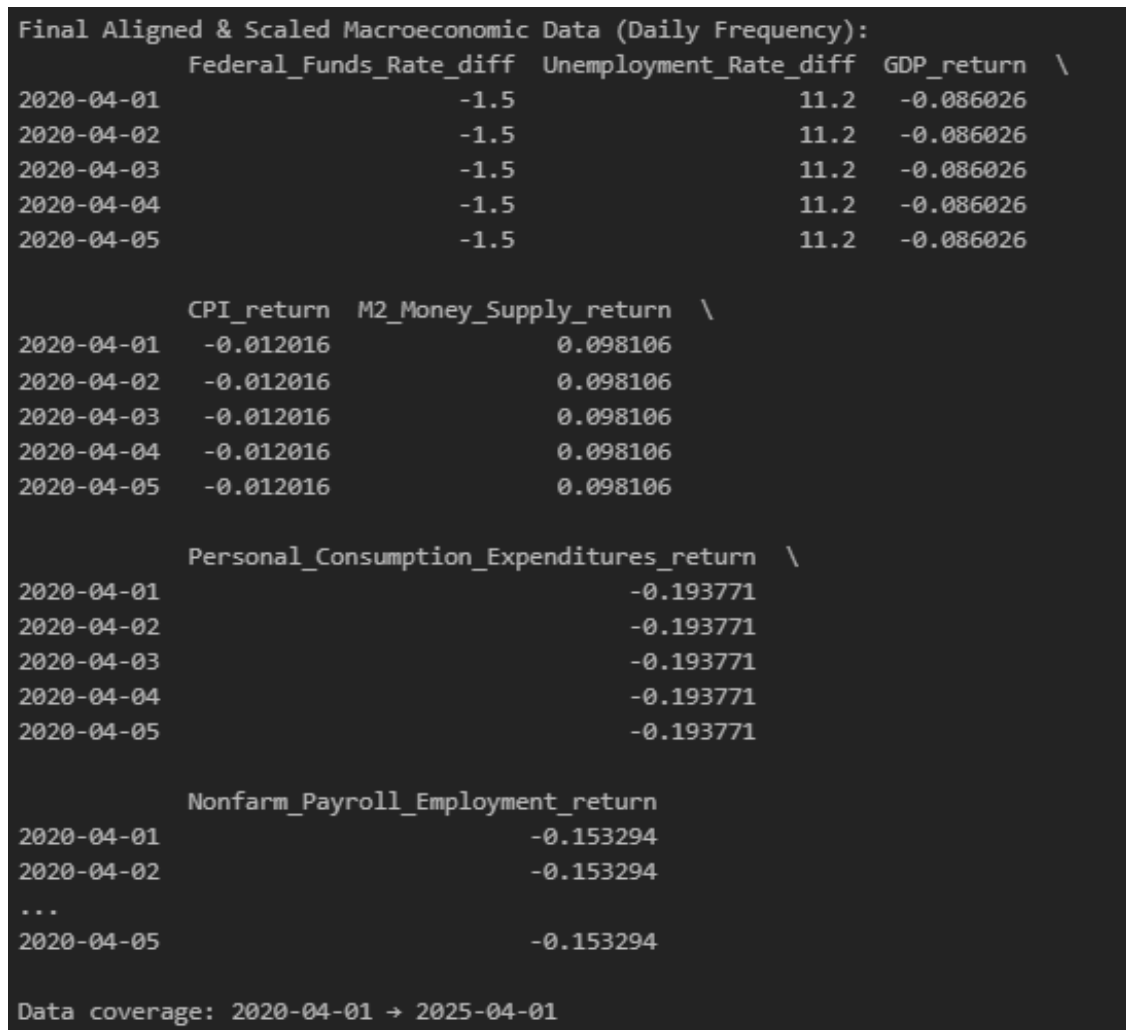


Figure 10: Macroeconomic Indicators Used in Analysis

✓ Daily Sentiment (final):		
	weighted_sentiment_vader	weighted_sentiment_textblob
date		
2020-01-10	5.375486	1.601786
2020-01-16	-3.995771	0.004525
2020-01-20	1.983299	0.369466
2020-01-24	0.000000	0.000000
2020-01-31	1.529621	0.599146

Figure 11: Daily Weighted Reddit Sentiment

Display first 50 rows of scraped Reddit data

reddit_df.head(50)

✓ 0.0s

Open 'reddit_df' in Data Wrangler

	🔍 ticker	📄 title	📄 text	📅 date	📊 # score	💬 # comments	🌐 url	🏠 subreddit
0	SPX	End of march bounce, thanks bears	spx gambol yesterday two hours bef	2020-04-01		114	21 https://www.reddit.com/gallery/1j08	WallStreetBets
1	SPX	I borrowed \$4.2M to invest. Should I	You might remember me from [4.5 y	2020-03-27		3183	1169 https://www.reddit.com/r/wallstreet	WallStreetBets
2	SPX	SPX DO: (Almost) Every Time SPX HI	***TLDR: 90% chance SPY goes to \$51	2020-03-27		140	85 https://www.reddit.com/r/wallstreet	WallStreetBets
3	SPX	Made 15k USD PRE MARKET SPX go		2020-03-24		245	39 https://www.reddit.com/gallery/1j0d	WallStreetBets
4	SPX	Somehow made money on both put.	Ignore those dips	2020-03-19		508	86 https://www.reddit.com/gallery/1j06	WallStreetBets
5	SPX	99% of gamblers quit before hitting	today's play that got me throug	2020-03-14		1381	139 https://www.reddit.com/gallery/1j05	WallStreetBets
6	SPX	Quick 8.5 bagger pre market SPX pig		2020-03-13		43	43 https://www.reddit.com/gallery/1j0a	WallStreetBets
7	SPX	I wonder what happened to this amt		2020-03-10		649	15 https://www.reddit.com/gallery/1j01	WallStreetBets
8	SPX	11k up in 15 minutes LFG	Held 3075 569 Puts 80 contracts Av	2020-03-10		41	15 https://www.reddit.com/gallery/1j0y	WallStreetBets
9	SPX	My portfolio got 100% tank	👉 new post: r.potato.madison 🍌	2020-03-04		424	10 https://i.redd.it/2zuaeeoone1.jpg	WallStreetBets

Mon

Figure 12: Distribution of Reddit Sentiment

Autocorrelation Functions

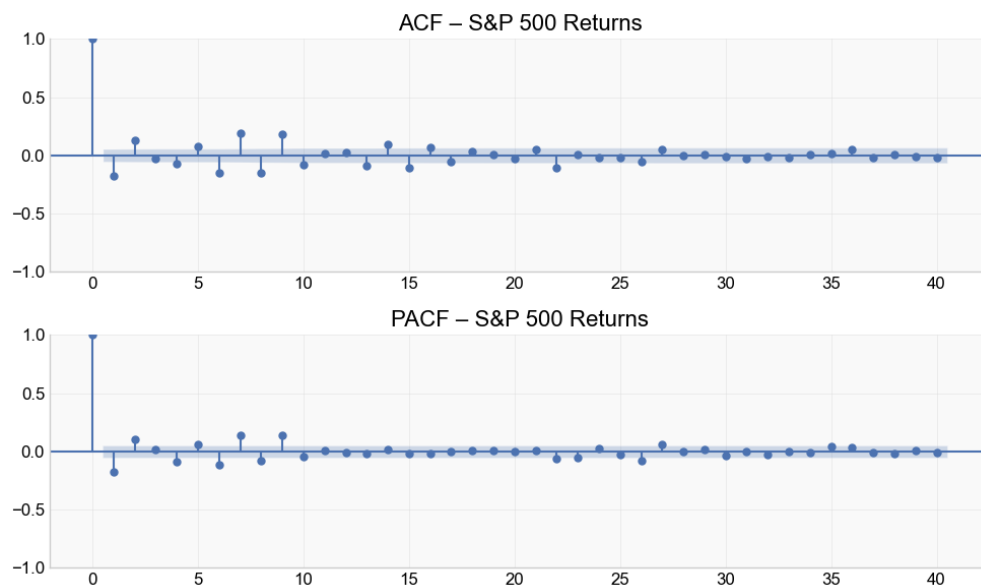


Figure 13: Autocorrelation Function: SPX Returns

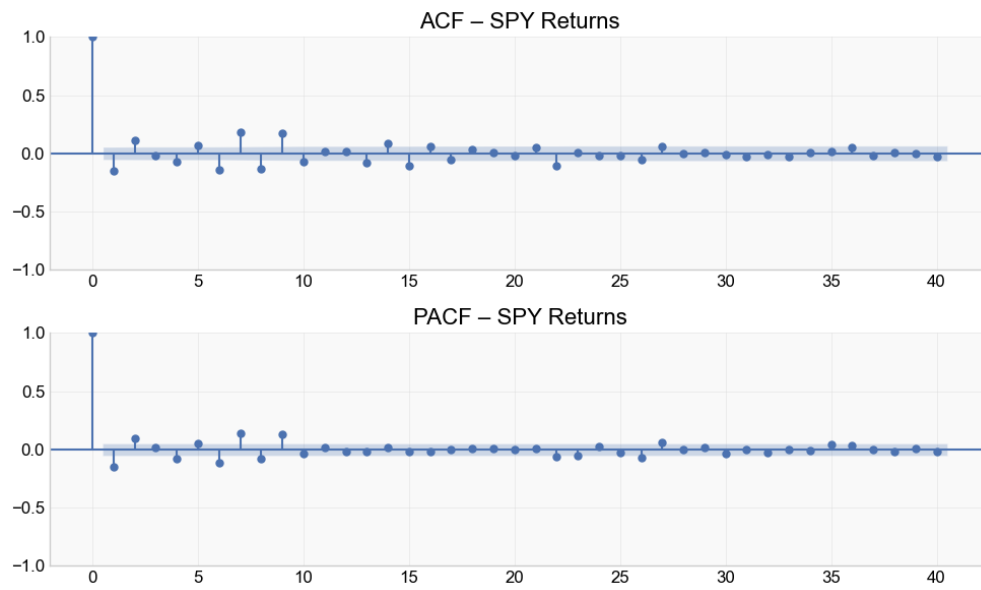


Figure 14: Autocorrelation Function: SPY Returns

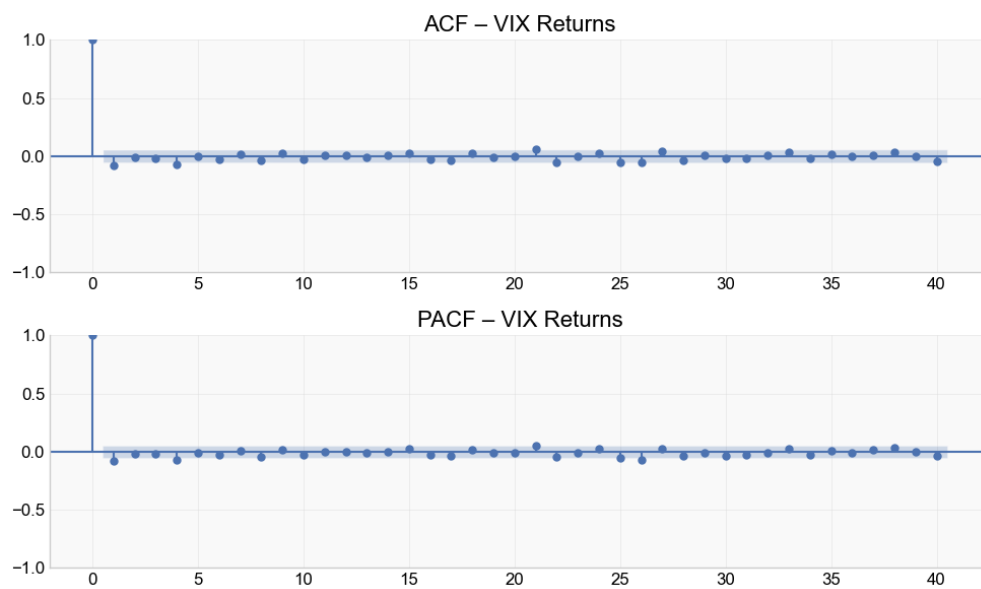


Figure 15: Autocorrelation Function: VIX Returns

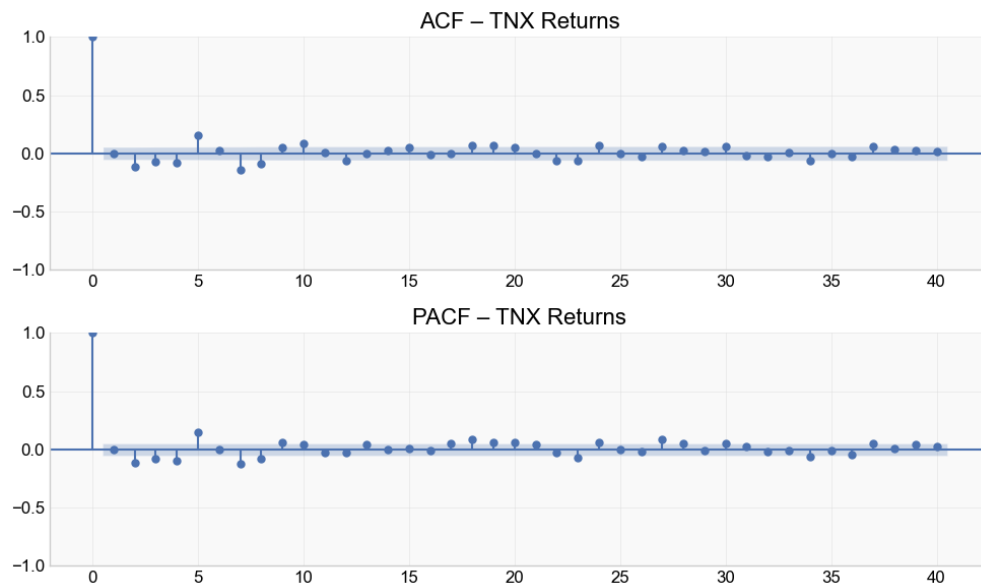


Figure 16: Autocorrelation Function: TNX Returns

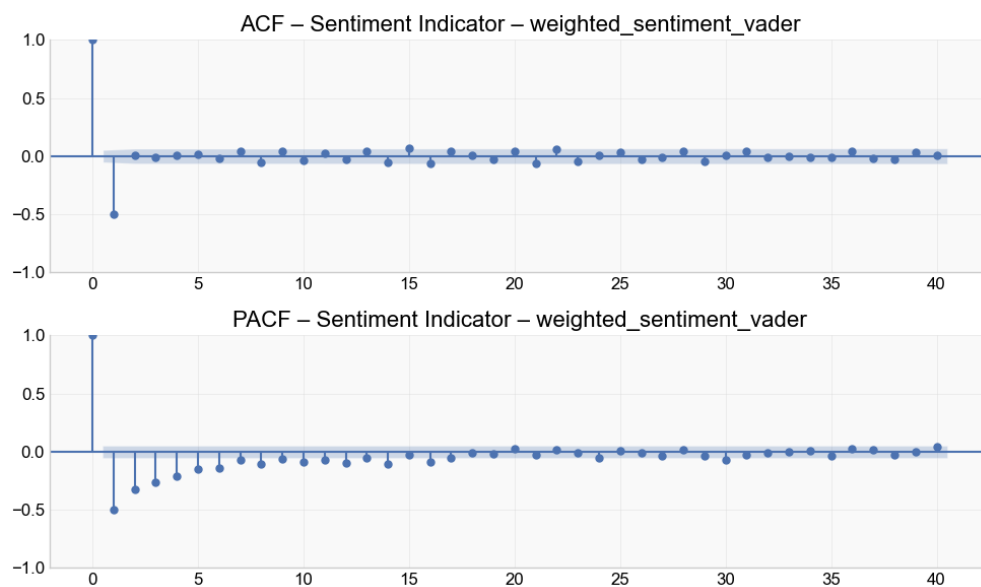


Figure 17: Autocorrelation of VADER Sentiment

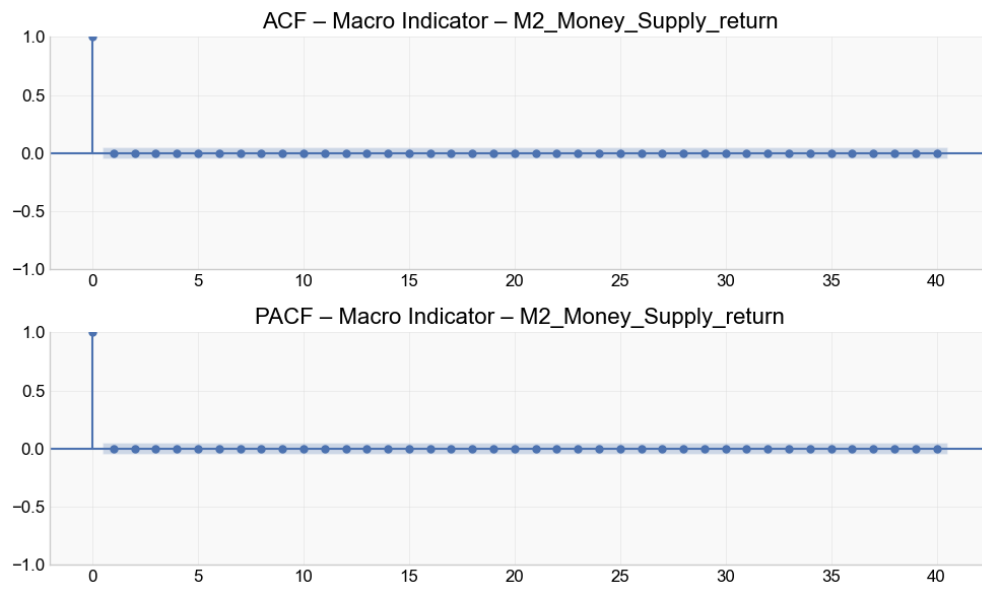


Figure 18: Autocorrelation of M2 Growth

GARCH Model Diagnostics

```

===== MODEL SUMMARY =====
                        AR - GARCH Model Results
=====
Dep. Variable:          Returns    R-squared:                0.031
Mean Model:             AR        Adj. R-squared:              0.030
Vol Model:              GARCH     Log-Likelihood:             4123.25
Distribution:           Normal    AIC:                     -8236.50
Method:                Maximum Likelihood    BIC:                     -8210.59
                                           No. Observations:        1316
Date:                  Mon, Dec 01 2025    Df Residuals:             1314
Time:                  16:56:53           Df Model:                  2
                                           Mean Model

=====
              coef    std err          t      P>|t|      95.0% Conf. Int.
-----
Const       9.4593e-04  2.605e-04      3.631  2.819e-04  [4.354e-04,1.456e-03]
Returns[1]  -0.1787  3.009e-02     -5.939  2.865e-09  [ -0.238, -0.120]
              Volatility Model
=====
              coef    std err          t      P>|t|      95.0% Conf. Int.
-----
omega       3.4647e-06  1.116e-13  3.105e+07      0.000  [3.465e-06,3.465e-06]
alpha[1]     0.1000  1.504e-03     66.487      0.000  [9.705e-02, 0.103]
beta[1]      0.8800  6.553e-03    134.287      0.000  [ 0.867, 0.893]
=====
Covariance estimator: robust

```

Figure 19: AR(1)–GARCH(1,1) Model Summary



Figure 20: Conditional Volatility from AR(1)–GARCH(1,1)

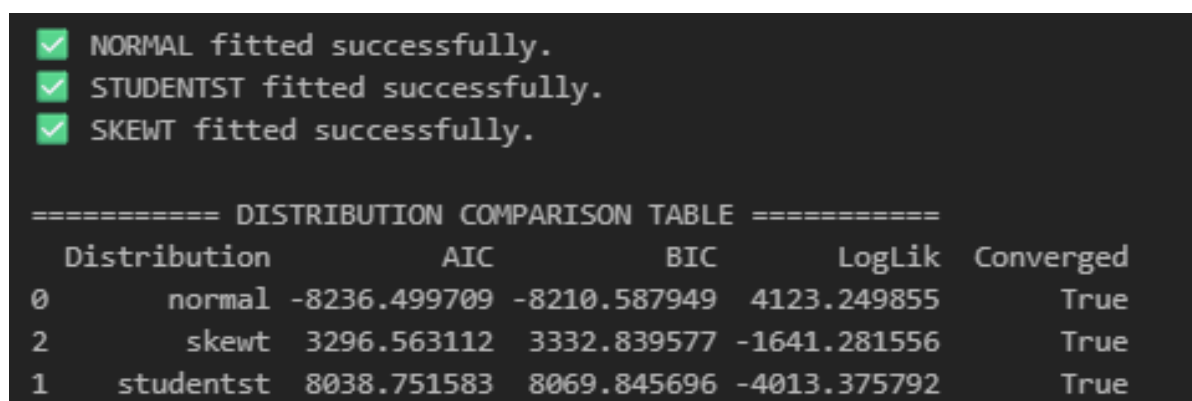


Figure 21: GARCH Innovation Distribution Selection (Normal vs Student- t vs Skew- t)

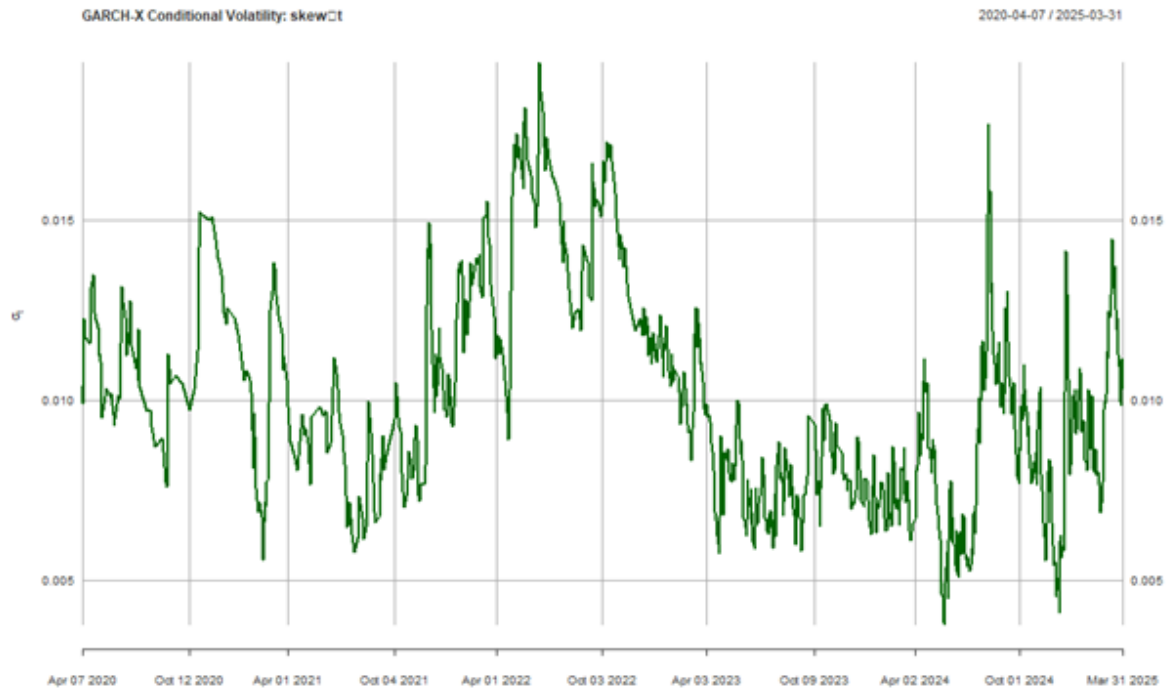


Figure 22: GARCH-X Conditional Volatility Across Distributions

Granger Causality Tests

```

=== GRANGER CAUSALITY TESTS ===
Federal_Funds_Rate_diff → Returns: lag 5, p = 0.1792
Unemployment_Rate_diff → Returns: lag 4, p = 0.0563
GDP_return → Returns: lag 4, p = 0.1476
CPI_return → Returns: lag 4, p = 0.2030
M2_Money_Supply_return → Returns: lag 1, p = 0.5781
Personal_Consumption_Expenditures_return → Returns: lag 4, p = 0.1694
Nonfarm_Payroll_Employment_return → Returns: lag 4, p = 0.0832
weighted_sentiment_vader → Returns: lag 5, p = 0.1164
weighted_sentiment_textblob → Returns: lag 4, p = 0.1356

=== SUMMARY ===
Macroeconomic causality summary:

```

	cause	effect	min_p_value	best_lag
1	Unemployment_Rate_diff	Returns	0.056304	4
6	Nonfarm_Payroll_Employment_return	Returns	0.083194	4
2	GDP_return	Returns	0.147609	4
5	Personal_Consumption_Expenditures_return	Returns	0.169409	4
0	Federal_Funds_Rate_diff	Returns	0.179194	5
3	CPI_return	Returns	0.202983	4
4	M2_Money_Supply_return	Returns	0.578128	1

```

Sentiment causality summary:

```

	cause	effect	min_p_value	best_lag
0	weighted_sentiment_vader	Returns	0.116446	5
1	weighted_sentiment_textblob	Returns	0.135555	4

```

✅ Combined dataset shape: (853, 13)
Columns available for Granger causality: ['SPX>Returns', 'SPY>Returns', 'VIX>Returns',
TNX>Returns: Stationary after differencing 1x.
Federal_Funds_Rate_diff: Stationary after differencing 1x.
CPI_return: Stationary after differencing 1x.
M2_Money_Supply_return: Stationary after differencing 2x.

Running Granger causality tests...

=====
Significant Causality Results
=====
VIX>Returns → SPX>Returns | p = 0.0904 | lag = 2
M2_Money_Supply_return → SPX>Returns | p = 0.0082 | lag = 2

```

Figure 23: Granger Causality Test Results

Deep Learning Model Results

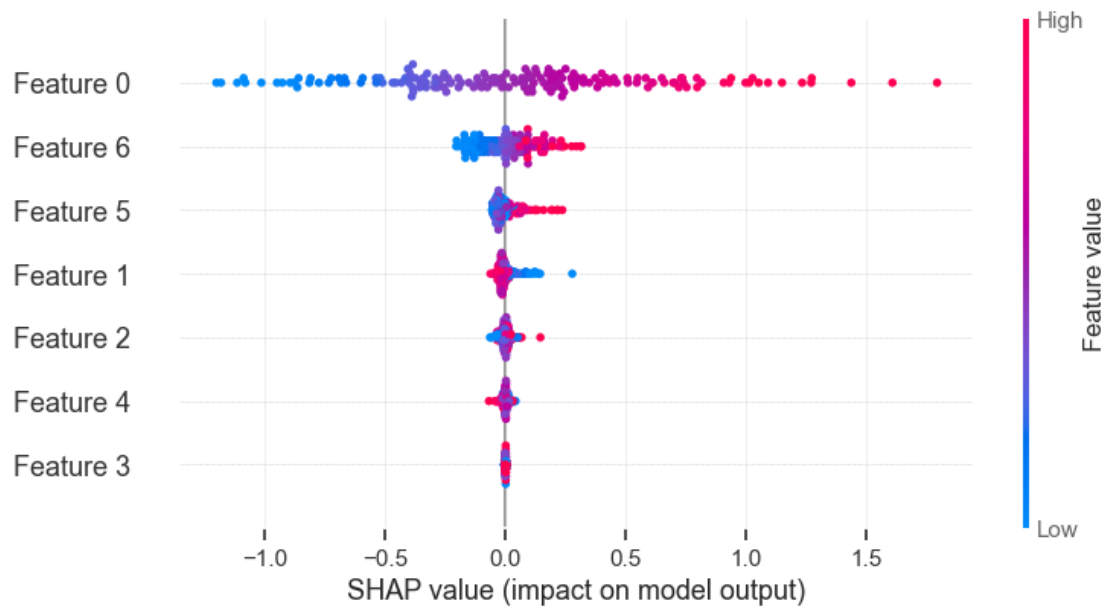


Figure 24: SHAP Values for DNN Volatility Forecasting Model



Figure 25: Deep Learning Forecast vs Realized Volatility

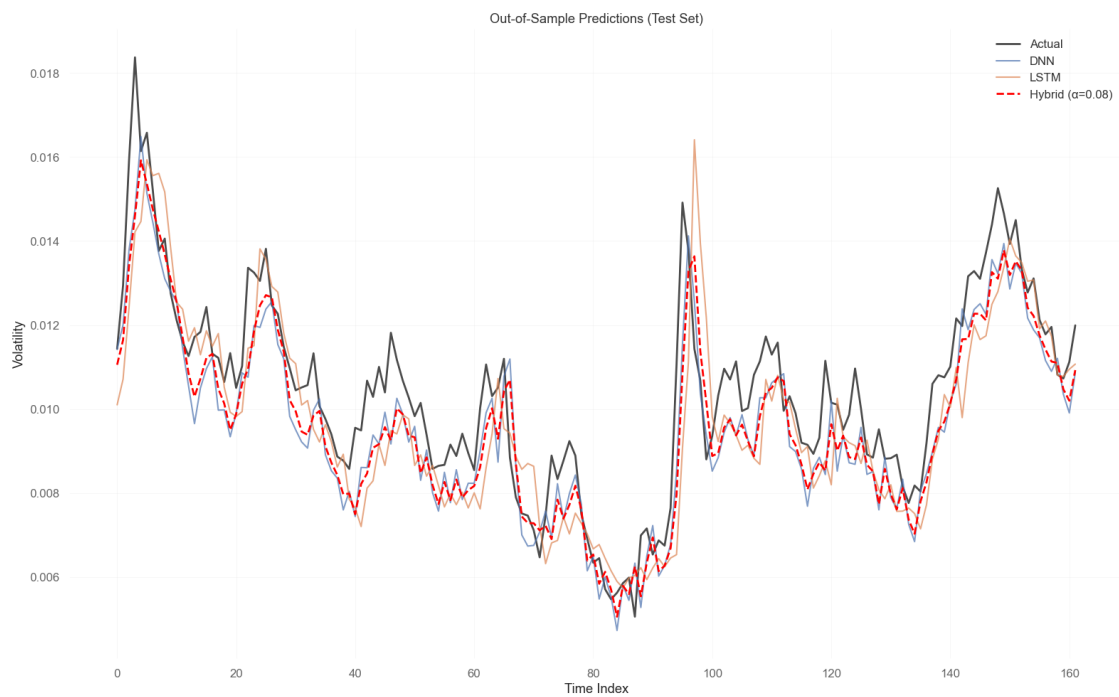


Figure 26: Volatility Forecasts Across Deep Learning Models

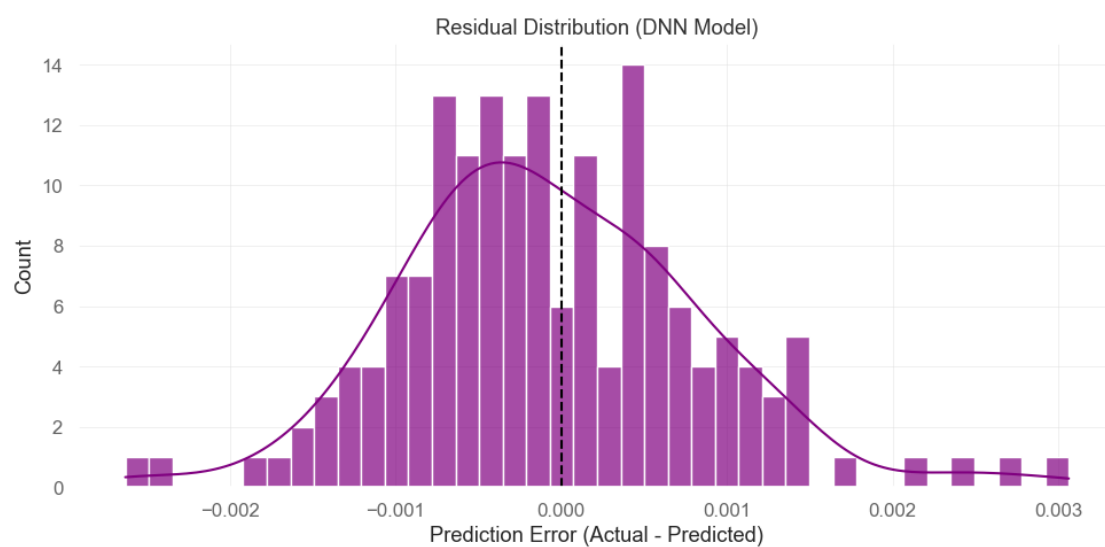


Figure 27: Residual Distribution: DNN Model

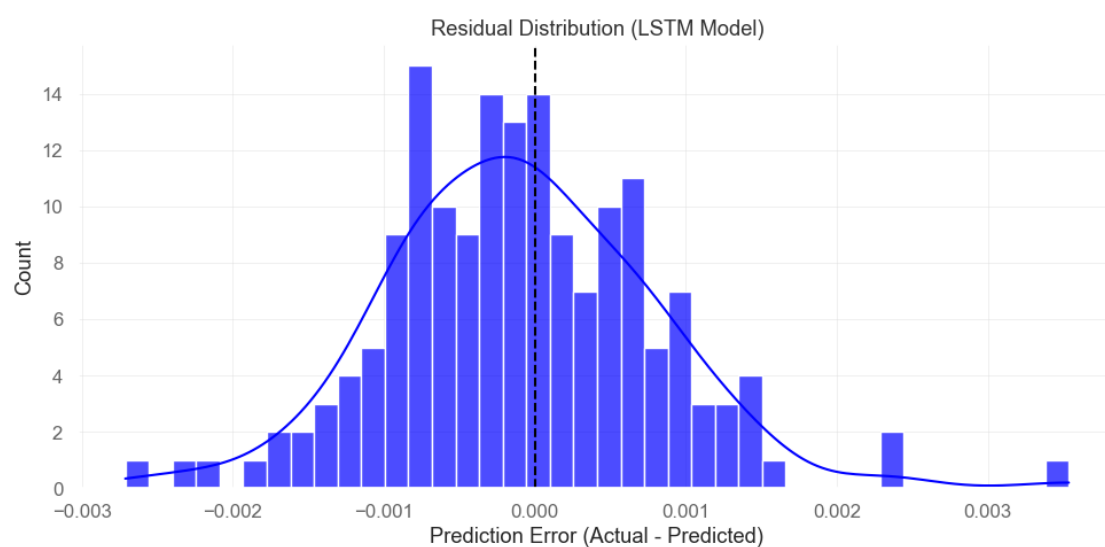


Figure 28: Residual Distribution: LSTM Model

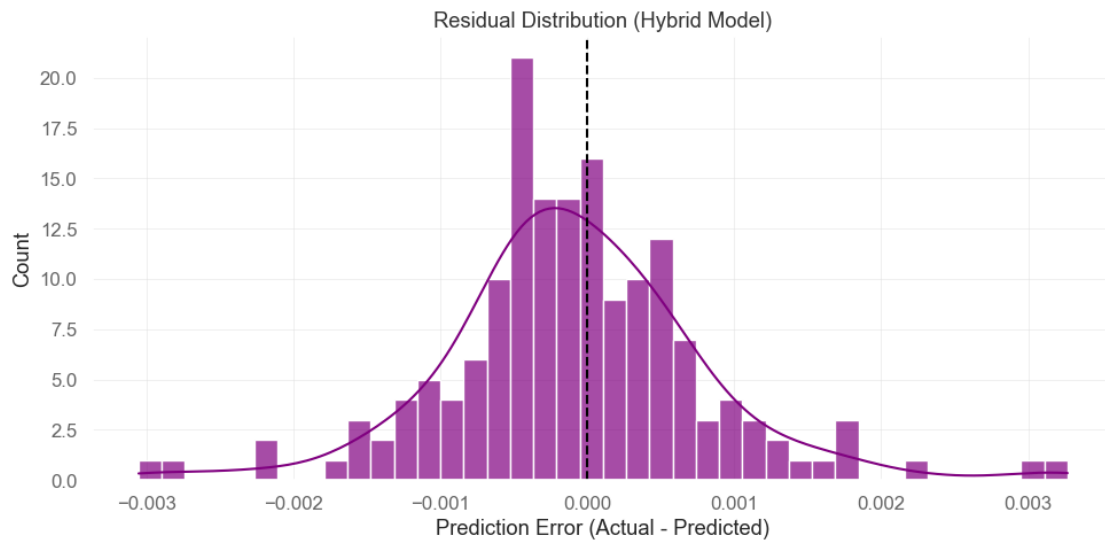


Figure 29: Residual Distribution: Hybrid Ensemble

```
=====
OUT-OF-SAMPLE PERFORMANCE (Test Set)
=====
```

	MAE	MSE	RMSE	R ²
DNN	0.000658	0.000001	0.000870	0.869151
LSTM	0.000996	0.000002	0.001348	0.685982
Hybrid ($\alpha=0.46$)	0.000696	0.000001	0.000938	0.848042

```
✓ Best model (lowest MAE): DNN
=====
```

Figure 30: Final Performance Comparison of Deep Learning Models